Reading the Market: News-Augmented Regime Discovery via Unsupervised Representation Learning

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***Abstract*—Hidden-Markov regime models for financial markets operate on price-derived features alone, so the regimes they recover are reactive: they change only after information has been absorbed by prices. We ask whether unsupervised representations of financial news yield regimes that are better separated and more persistent than a price-only baseline. On 6,262 trading days of the S&P 500 (2000–2024) we compare 18 clustering pipelines spanning price-only, news-only, and price-plus-news feature stacks, built from four text representations (sentiment scalars, latent Dirichlet allocation, BERTopic, and FinBERT) and two news corpora (firm-level press releases and aggregate macro news). A held-out grid search over 180 configurations then tunes the representation, dimensionality reduction, clustering method, and number of regimes. News augmentation raises the silhouette score by 34% over the price-only hidden Markov model (0.202 to 0.270) while keeping regimes inside an interpretable multi-month persistence band. Aggregate macro news outperforms firm-level press releases, and the tuned configuration (BERTopic + PCA-95% + k-means, $K = 5$) reaches a silhouette of 0.279 at a 202-day mean regime duration. Adding price features back on top stabilizes regimes further (224 days) at negligible cost to separation.**

***Index Terms*—unsupervised learning, market regimes, representation learning, clustering, financial news**

I. Introduction

A *market regime* is a persistent state of the financial market characterized by distinct return and risk behavior, such as a calm bull market or a high-volatility crisis. The standard tool for recovering regimes is the hidden Markov model (HMM) of Hamilton [1], which treats the regime as a latent discrete state driving observed price features such as returns and volatility [2]. Because these models read only from prices, the regimes they discover are reactive by construction: a regime boundary can move only once new information has already changed returns or volatility.

Financial news is a natural candidate for a complementary, possibly earlier, signal. Prior work shows that news sentiment and news topics carry information about returns and aggregate risk [3], [4], [5]. That work is mostly supervised or forecasting-oriented. We instead ask an *unsupervised* question: without any labeled regimes, do representations of news produce a regime *structure* that is better separated and more persistent than the structure recovered from prices alone?

We study this question as a controlled comparison of clustering pipelines. The contributions are:

- A systematic comparison of four text representations (sentiment scalars, LDA, BERTopic, FinBERT) crossed with three clustering algorithms (Gaussian mixtures, HMMs, k-means) for unsupervised regime discovery, against a price-only HMM baseline.
- A comparison of two news corpora at different granularity: firm-level press releases and aggregate macro news.
- A four-axis evaluation protocol that scores each pipeline on statistical fit, clustering geometry, temporal stability, and financial interpretability, rather than on any single number.
- A practical finding: full-covariance HMMs collapse to an absorbing state on high-dimensional fused features, which a switch to diagonal covariance fixes.

II. Related Work

Regime models. Hamilton [1] introduced the Markov-switching model for nonstationary economic series; the approach is now standard for markets [2]. These models use price- and macro-derived observables. We keep the same latent-state view but ask whether text representations improve the recovered states under unsupervised evaluation.

News and markets. Tetlock [3] links media sentiment to returns; Glasserman and Mamaysky [4] build an "unusual news" indicator that forecasts market stress; and Bybee et al. [5] show that news topics track the business cycle. These studies predict a known target (returns, volatility, or recession dates). Our setting has no target: we cluster, then ask whether the clusters are geometrically and economically coherent.

Text representations. We use latent Dirichlet allocation [6], BERTopic [7], and FinBERT [8] embeddings [9] as off-the-shelf encoders, and study how each behaves as input to clustering rather than proposing a new encoder.

III. Data

Market features. We build five daily features for the S&P 500 over 2000–2024 from CRSP and FRED: the daily return, 21-day realized volatility, the VIX, the term spread (10Y minus 2Y Treasury yield), and the credit spread (Baa minus Aaa corporate yield). After aligning all sources we retain 6,262 trading days.

News corpora. We pair the market series with two RavenPack news corpora at two levels of granularity. The *firm-level*

corpus (Dow Jones press releases, "DJPR") is filtered to US entities (`entity_type = COMP`) with relevance ≥ 75. The *macro* corpus (Dow Jones Global Macro) carries about 24 million rows across 27 yearly partitions and captures aggregate attention to rates, commodities, currencies, and global events; it begins in 2000. For each corpus we aggregate to daily series of sentiment statistics (mean and standard deviation of composite sentiment, article count, positive/negative fractions, and net sentiment).

IV. Methods

Each pipeline maps raw inputs to a daily sequence of regime labels through five stages: assemble features, smooth, represent text, reduce dimensionality, and cluster. We write $\mathbf{x}_t \in \mathbb{R}^d$ for the assembled feature vector on trading day $t = 1, \ldots, T$ (with $T = 6{,}262$ after alignment) and $z_t \in \{1, \ldots, K\}$ for the regime label a pipeline assigns to that day.

A. Representations and smoothing

We compute four representations of each news corpus. *Scalar* is the five-dimensional vector of daily sentiment statistics. *LDA* [6] gives a $K \in \{15, 20\}$-dimensional topic distribution per day. *BERTopic* [7] yields about 20 topic weights from contextual embeddings, and *FinBERT* [8] provides a 768-dimensional embedding averaged to a daily vector. Daily news is noisy, so before clustering we replace each feature stream by a trailing rolling mean of window w,

$$\tilde{\mathbf{x}}_t = \frac{1}{w} \sum_{j=0}^{w-1} \mathbf{x}_{t-j}, \tag{1}$$

with w set per representation (35 days for scalar features, 30 for LDA, 21 for BERTopic, and 10 for FinBERT); topic representations are already daily aggregates and need less smoothing than the raw scalars.

B. Feature stacks and reduction

We enumerate three pathways: *price-only* (the five market features), *news-only* (one or both corpora in one text representation), and *hybrid* (price plus news). Crossing pathways, representations, and corpora gives the 18 main pipelines. Before clustering we optionally reduce the assembled stack with principal component analysis, keeping the leading components that explain 95% of the variance, or with UMAP [10] to ten dimensions.

C. Clustering models

We cluster the daily feature vectors with three models. The Gaussian mixture model treats each day as an independent draw from

$$p(\mathbf{x}_t) = \sum_{k=1}^{K} \pi_k \, \mathcal{N}(\mathbf{x}_t; \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k), \tag{2}$$

and assigns the regime with the largest posterior responsibility, $z_t = \arg\max_k \pi_k \, \mathcal{N}(\mathbf{x}_t; \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$. The hidden Markov model adds a first-order dependence between consecutive days through a $K \times K$ transition matrix $\mathbf{A}$,

$$p(\mathbf{x}_{1:T}, z_{1:T}) = \pi_{z_1} \prod_{t=2}^{T} A_{z_{t-1} z_t} \prod_{t=1}^{T} \mathcal{N}(\mathbf{x}_t; \boldsymbol{\mu}_{z_t}, \boldsymbol{\Sigma}_{z_t}), \tag{3}$$

which is what lets it produce temporally persistent regimes. We initialize $\mathbf{A}$ with a sticky diagonal $A_{ii} = 0.85$, fit the parameters by expectation–maximization (300 iterations, keeping the best of five random seeds by log-likelihood), and decode the most likely path $z_{1:T}$ with the Viterbi algorithm. The k-means model ignores time and minimizes the within-cluster sum of squares $\sum_t \|\mathbf{x}_t - \mathbf{c}_{z_t}\|^2$ over centroids $\mathbf{c}_k$.

D. Choosing K and the covariance

For the probabilistic models we choose the regime count $K \in \{3, 4, 5\}$ by the Bayesian information criterion for the mixture and by log-likelihood for the hidden Markov model; for k-means we test the same three values directly. A full covariance $\boldsymbol{\Sigma}_k$ carries $O(d^2)$ free parameters per regime. On fused stacks with $d \geq 10$ this overfits and the hidden Markov model collapses to an absorbing state, with one regime absorbing almost all days and the mean duration running away to 894 days. Constraining the emissions to a diagonal covariance whenever $d \geq 10$ recovers multi-month regimes, and we adopt it as the default.

V. Evaluation Protocol

A clustering has no ground-truth labels, so we score each pipeline on four axes rather than a single number.

A. Statistical fit

For the probabilistic models we report the maximized log-likelihood $\hat{L}$ and the Bayesian and Akaike information criteria,

$$\mathrm{BIC} = -2 \log \hat{L} + m \log T, \qquad \mathrm{AIC} = -2 \log \hat{L} + 2m, \tag{4}$$

where m is the number of free parameters. Both reward fit and penalize model size, so they let us compare regime counts on the same scale.

B. Clustering geometry

We measure regime separation with the silhouette coefficient [11]. For day i, let $a(i)$ be its mean distance to the other days in its own regime and $b(i)$ the smallest mean distance to any other regime; then

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i),\, b(i)\}} \in [-1, 1], \tag{5}$$

and we report the average $S = \frac{1}{T} \sum_i s(i)$, where higher is better. We also report the Davies–Bouldin index [12],

$$\mathrm{DBI} = \frac{1}{K} \sum_{k=1}^{K} \max_{l \neq k} \frac{\sigma_k + \sigma_l}{\|\mathbf{c}_k - \mathbf{c}_l\|}, \tag{6}$$

where σ_k is the mean distance of regime-k days to their centroid $\mathbf{c}_k$, and lower is better. Unless noted, both run on a common standardized feature space so that pipelines are comparable.

C. Temporal stability

The regime sequence partitions time into maximal runs of a constant label; we report the mean run length in days and treat the band of 60 to 365 days as interpretable, since shorter regimes flicker too fast to name and longer ones are too coarse to be useful. To check robustness we also recluster within a rolling 1250-day window and compare the local labels to the full-sample labels with the adjusted Rand index, reporting its mean.

D. Financial interpretability

For each regime k we compute the annualized Sharpe ratio $\sqrt{252}\,\bar{r}_k/\mathrm{sd}(r_k)$ over the days it covers, along with the maximum drawdown and the average VIX, and report the best regime's Sharpe as a summary of whether the discovered states separate good and bad market conditions.

VI. Experiments and Results

A. Baseline

The price-only HMM recovers five regimes with a 95-day mean duration and a clean crisis state: one regime carries an average VIX of 41 and a -51% maximum drawdown, while the calmest regime has a Sharpe of 2.0 and an average VIX of 13. The baseline therefore separates market conditions well by financial measures, but its short 95-day regimes change quickly because they react to price moves.

B. Does news help?

Table I compares the baseline with the news-augmented pipelines. News augmentation raises the silhouette from 0.202 (price-only HMM) to 0.270 (macro sentiment HMM), a 34% improvement, and lowers the Davies–Bouldin index. The improvement is not specific to one representation: the scalar, LDA, and BERTopic representations on the macro corpus all beat the price baseline on silhouette. Fig. 2 shows the silhouette/Sharpe trade-off across all main pipelines: text-only pipelines tend to cluster at high silhouette, while hybrid pipelines keep the baseline's higher Sharpe. The price baseline does retain the highest rolling adjusted Rand index (0.447), so news improves separation and persistence rather than robustness to the estimation window.

C. Aggregate news beats firm-level news

A higher silhouette is not enough on its own. The DJPR sentiment HMM has the highest silhouette in the table (0.284) but a 44-day mean duration, so its regimes flicker too fast to interpret as market states. The macro sentiment HMM, by contrast, lands inside the interpretable band at 174 days while still beating the baseline on silhouette. Aggregate attention to rates, commodities, and global events is a steadier regime signal than the firm-level press-release stream, whose day-to-day sentiment is dominated by idiosyncratic company news.

TABLE I

Master evaluation. Arrows give the preferred direction. ⋆ marks configurations selected by the grid search of Section VI-D. Silhouette and DBI for the two tuned pipelines are measured in the reduced (PCA-95%) space in which they were selected; the other rows use the common standardized space of Section V.

Pipeline	Sil. ↑	DBI ↓	Dur. (d)	ARI ↑	Sharpe
Price-only HMM (baseline)	0.202	1.48	95	0.447	2.02
DJPR sentiment HMM	0.284	1.26	44	0.383	0.80
Macro sentiment HMM ⋆	0.270	1.27	174	0.239	0.65
DJPR topic-vector HMM	0.261	1.49	70	0.278	0.64
Dual-corpus sentiment fusion	0.245	1.42	49	0.268	1.00
Tuned news fusion ⋆	0.279	0.95	202	0.31	0.78
Tuned full stack ⋆	0.272	0.92	224	0.33	1.21

D. Tuning the pipeline

We then run a grid search over 180 configurations (two feature stacks $\times$ 90 settings) along four axes: text representation (LDA-15, LDA-20, BERTopic, and FinBERT under PCA-20, PCA-50, or UMAP-10), post-stack dimensionality reduction (none, PCA-95%, UMAP-10), clustering algorithm (GMM, HMM, k-means), and K. We select the configuration with the highest silhouette *subject to* a mean duration in the 60–365-day band. This constraint matters: the highest raw silhouettes (0.55–0.66) all come from FinBERT + UMAP configurations whose regimes last 270–1250 days, well past the interpretable range, and whose silhouette is measured in the very UMAP space that UMAP optimizes for separation, so it is not comparable to the PCA or raw-space scores. The best configuration within the band is BERTopic + PCA-95% + k-means with $K = 5$, which reaches a silhouette of 0.279 at a 202-day mean duration.

E. Price stabilizes news-driven regimes

Adding the five market features on top of the tuned news fusion (the "full stack" row) increases the mean regime duration from 202 to 224 days at negligible cost to silhouette (0.279 to 0.272) and raises the best-regime Sharpe from 0.78 to 1.21. Price acts as a low-frequency anchor that steadies the news-driven regimes against narrative noise.

F. Regime characterization

Fig. 1 overlays the regime sequences on the cumulative S&P 500. The news-augmented pipelines recover the same crisis structure as the price baseline: the dot-com bust, the global financial crisis, and the COVID crash all sit at regime boundaries, but the news-driven segments are smoother and longer. The tuned news fusion, for example, places its 2000–2004 regime at an average VIX of 25 and a negative Sharpe, and its post-2021 regime in the rate-hike era, matching the economic narrative without ever seeing prices.

VII. Discussion

The results support a specific, limited claim: unsupervised news representations recover regimes that are better separated

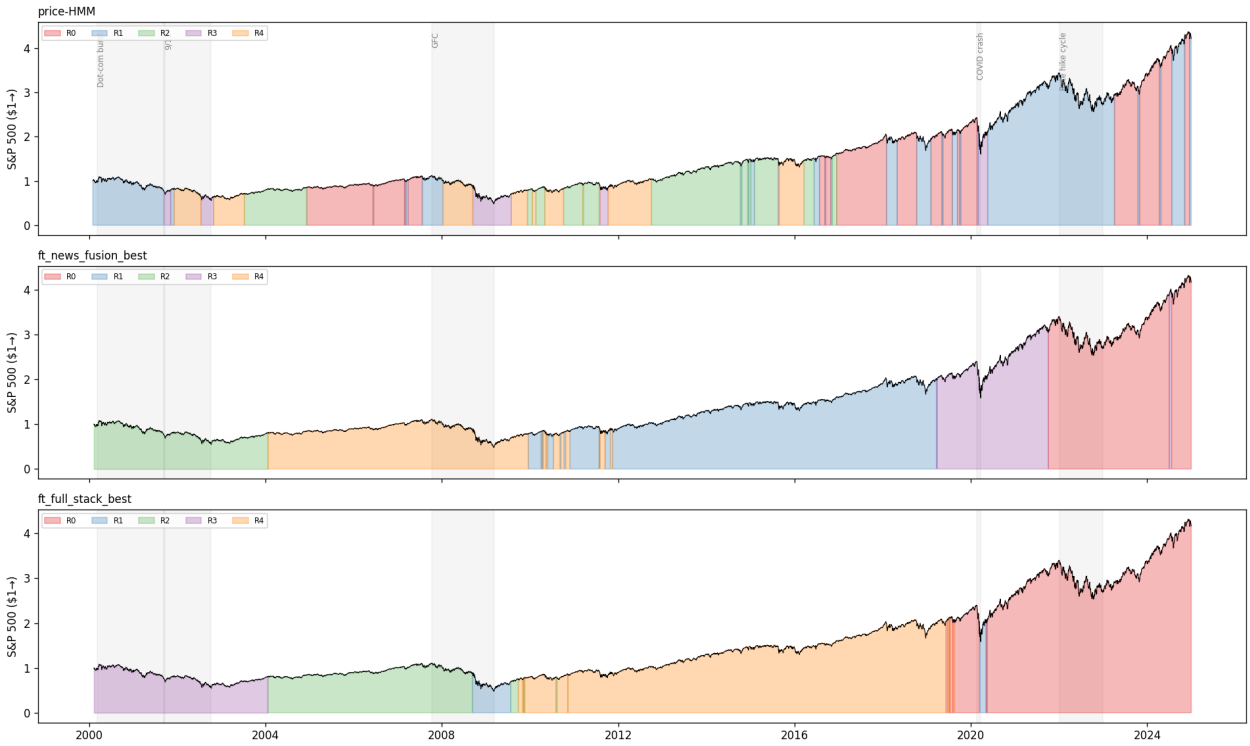


Fig. 1. Regime timelines on the cumulative S&P 500 ($1 growth). Top: price-only HMM baseline (95-day mean duration). Middle: tuned news fusion (BERTopic + PCA-95% + k-means, $K = 5$; 202-day duration). Bottom: tuned full stack (+ market features; 224-day duration). Shaded vertical bands mark known crises. News-augmented regimes capture the same crisis dynamics with shifted, smoother boundaries.

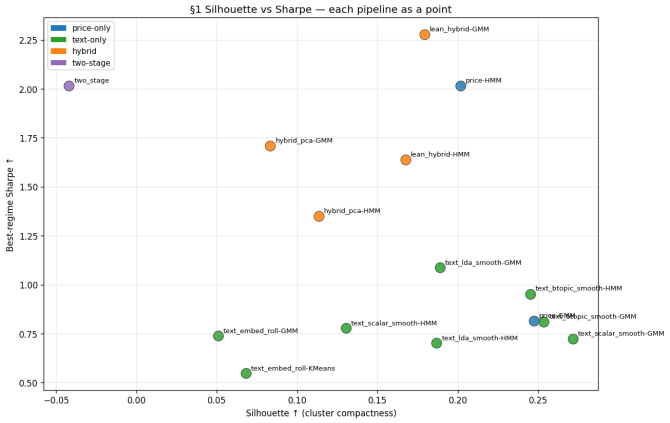


Fig. 2. Silhouette versus best-regime Sharpe for the main pipelines, colored by feature family. Text-only pipelines reach the highest silhouettes; hybrid pipelines retain the price baseline's higher Sharpe. Silhouette is computed on the common standardized space.

(higher silhouette, lower DBI) and, on the macro corpus, more persistent than the price-only baseline, while still aligning with known crises and with sensible per-regime risk. They do *not* show that news leads prices; establishing that would require an explicit lead–lag test, which we leave to future work. The tuning experiment also makes a measurement caveat concrete: silhouette computed in a reduced space, especially a UMAP space, is not comparable across spaces, and a high silhouette paired with a multi-year regime is degenerate rather than good. Reporting clustering geometry alongside a persistence band keeps the comparison honest.

VIII. Limitations and Future Work

The evaluation is in-sample: every pipeline is fit on the full 2000–2024 window. A walk-forward out-of-sample check using a persisted HMM and `predict()` on a held-out post-2020 window would test stability under true generalization. We also report point estimates of silhouette without a bootstrap confidence interval, so pairwise differences between close pipelines are not formally significant. The macro corpus begins in 2000, so we cannot probe earlier regimes. Promising extensions are a replication of the unusual-news indicator of Glasserman and Mamaysky [4] as an external benchmark, and changepoint-aware adaptive smoothing in place of fixed windows.

Author Contributions

This is a solo project. Y. Huang designed the study, implemented the full pipeline and evaluation, ran all experiments, and wrote the report.

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